

<page_start no = 1>

<sol_start id=1>

1. (3) 3 : 2

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<sol_start id=2>

2. (1) ₹ 360

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<sol_start id=3>

3. (1) Convergent lens

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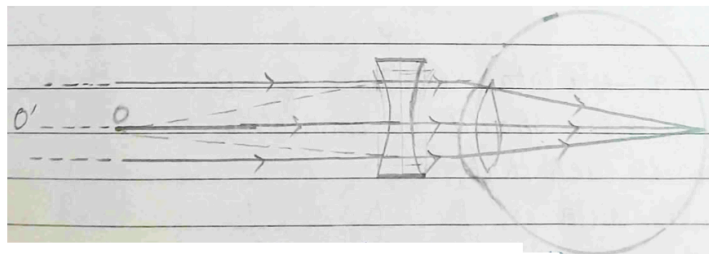
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4. (2) Common defect of vision

(1) Myopia

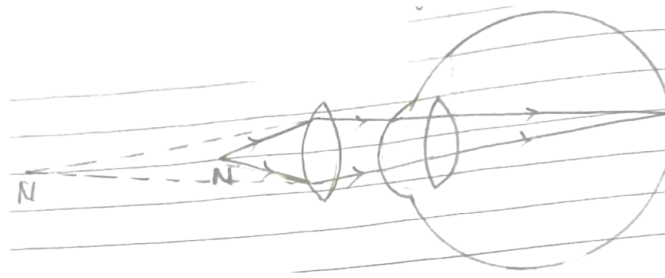
(2) Hypermetropia

Myopia can be corrected by using concave lens



Correction for Myopia

Hypermetropia can be corrected by using convex lens



Correction for Hypermetropia

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5.(i) Reading of Ammeter will be the total current flowing in the circuit.

So, now, equivalent resistance = $8 + 5 + 12 = 25\Omega$

Total voltage = $2V + 2V + 2V$

= $6V$

$$\text{Current} = \frac{V}{R} = \frac{6}{25} = 0.24A$$

$\therefore$ Reading of Ammeter = $0.24A$

Reading of voltmeter will be the voltage of across 12Ω resistor

$$\begin{aligned} \text{Voltage across } 12\Omega \text{ resistor} &= \frac{24}{100} \times 12 \\ &= \frac{72}{25} = 2.88V \end{aligned}$$

$\therefore$ Reading of voltmeter = $2.88V$

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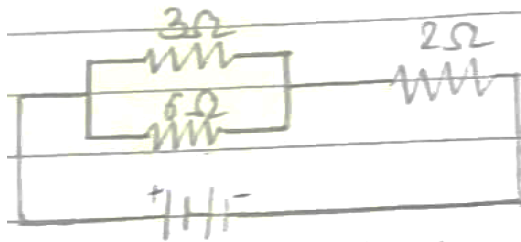
- (ii) To give a total resistance of 1Ω , all the resistors should be connected in parallel, such that,

$$\begin{aligned}\text{Equivalent Resistance, } \frac{1}{R_{eq}} &= \frac{1}{2} + \frac{1}{3} + \frac{1}{6} \\ &= \frac{1}{1}\Omega \Rightarrow R_{eq} = 1\Omega\end{aligned}$$

To give a total resistance of 4Ω , 3Ω and 6Ω resistors need to be connected in parallel and 2Ω resistor should be connected in series with the combination of 3Ω and 6Ω resistors, such that,

$$\frac{1}{6} + \frac{1}{3} = \frac{1+2}{6} = \frac{3}{6} \Rightarrow R_{eq} = 2\Omega$$

Now, $2\Omega + 2\Omega = 4\Omega$, which is the required total resistance.



Diagrammatic representation of 4Ω total resistance.

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<sol_start id=6>

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- 6.(i) (1) For point of myopic eye, $v = -80\text{ cm}$

$$u = -\infty$$

f, focal length = ?

Using lens formula, we get,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{-1}{80} - \left(\frac{-1}{\infty} \right) = \frac{1}{f}$$

Now, $f = -0.8 \text{ m}$

Power, $P = \frac{1}{f} = \frac{-1}{8} = -1.25 \text{ D}$

Nature of lens = Divergent, which is a concave lens.

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- (ii) Power of accommodation is the ability of eye lens to adjust its focal length, this ability of eye lens is controlled by ciliary muscles when the ciliary muscles contract, then the focal length of eye lens is less and when ciliary muscles relax, then its focal length of crystalline eye lens is more.

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- (iii) Role of iris :

(1) Controls the amount of light to be entered in the eye.

(2) Determines the colour of eye.

(3) Controls the size of pupil.

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